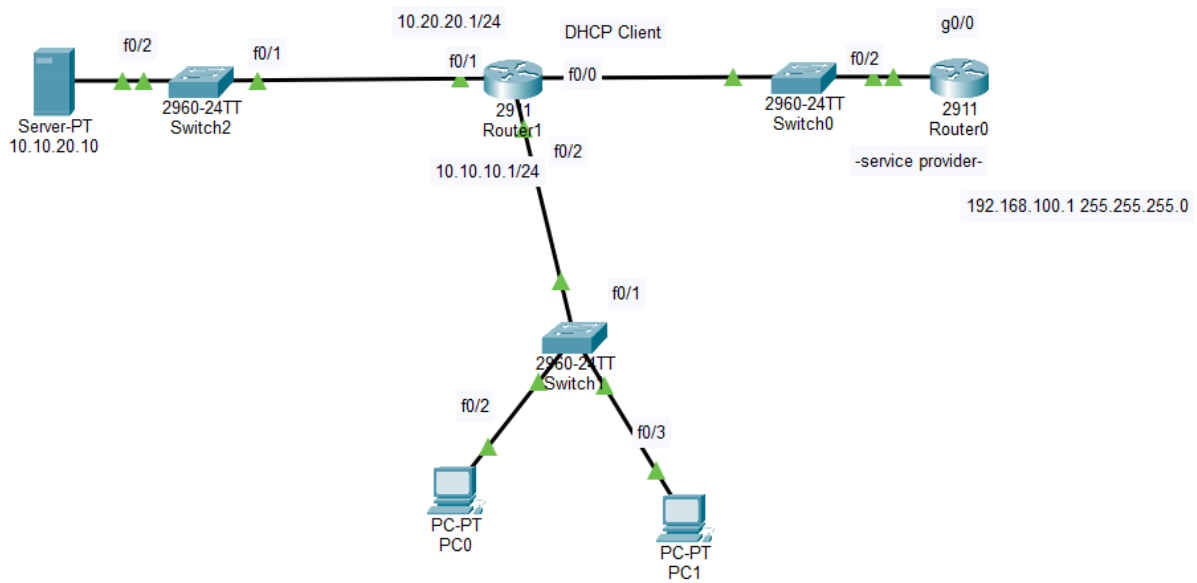




```
Router(config-if)#do show ip interface brief
Interface                IP-Address      OK? Method Status        Protocol
GigabitEthernet0/0       unassigned      YES DHCP    up            up
GigabitEthernet0/1       10.20.10.1      YES manual  up            up
GigabitEthernet0/2       10.10.10.1      YES manual  up            up
Vlan1                    unassigned      YES unset   administratively down down
Router(config-if)#
```

PC0
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
IP Configuration
DHCP Static
IPv4 Address 10.10.10.11
Subnet Mask 255.255.255.0
Default Gateway 10.10.10.1
DNS Server 10.20.10.10
IPv6 Configuration
Automatic Static
IPv6 Address
Link Local Address FE80::202:17FF:FEE1:7232
Default Gateway
DNS Server

PC1
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
IP Configuration
DHCP Static DHCP request successful.
IPv4 Address 10.10.10.12
Subnet Mask 255.255.255.0
Default Gateway 10.10.10.1
DNS Server 10.20.10.10
IPv6 Configuration
Automatic Static
IPv6 Address
Link Local Address FE80::201:96FF:FEAD:D500
Default Gateway
DNS Server

```
Router#show ip dhcp binding
```

IP address	Client-ID/ Hardware address	Lease expiration	Type
10.10.10.11	0002.17E1.7232	--	Automatic
10.10.10.12	0001.96AD.D500	--	Automatic

```
Router#
```

```
Router#show ip interface g0/1
GigabitEthernet0/1 is up, line protocol is up (connected)
  Internet address is 10.20.10.1/24
  Broadcast address is 255.255.255.255
  Address determined by setup command
  MTU is 1500 bytes
  Helper address is 10.20.10.10
  Directed broadcast forwarding is disabled
  Outgoing access list is not set
  Inbound access list is not set
  Proxy ARP is enabled
  Security level is default
  Split horizon is enabled
  ICMP redirects are always sent
  ICMP unreachable are always sent
  ICMP mask replies are never sent
  IP fast switching is disabled
  IP fast switching on the same interface is disabled
  IP Flow switching is disabled
  IP Fast switching turbo vector
  IP multicast fast switching is disabled
  IP multicast distributed fast switching is disabled
  Router Discovery is disabled
  IP output packet accounting is disabled
  IP access violation accounting is disabled
  TCP/IP header compression is disabled
  RTP/IP header compression is disabled
  Probe proxy name replies are disabled
  Policy routing is disabled
  Network address translation is disabled
  BGP Policy Mapping is disabled
  Input features: MCI Check
  WCCP Redirect outbound is disabled
  WCCP Redirect inbound is disabled
  WCCP Redirect exclude is disabled
```

DHCP clients initially send DHCP Discover messages as broadcasts because they do not yet have an IP address or know the location of the DHCP server. Routers normally do not forward broadcasts between different networks. Therefore, when the DHCP server is on a different subnet, the router needs an ip helper-address configured on the client-facing interface to relay the DHCP request to the remote DHCP server.